

03 PMF AND CDF (notes p063-p087)

1. The probability mass function, pmf (p064)

$p(x) = P(X = x) = P(\{s \text{ in } S : X(s) = x\})$
 "for every possible value x , the probability of observing it"

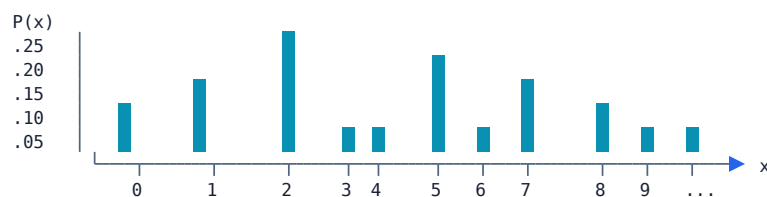
Two conditions (necessary AND sufficient):

- (i) $p(x) \geq 0$ every probability is non-negative
- (ii) $\sum p(x) = 1$ the probabilities over ALL possible x sum to 1

"The probability mass function of a discrete random variable describes the behaviour of the variable."

2. Worked: gas station pumps (p065)

$x : 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6$
 $p(x) : .05 \ .10 \ .15 \ .25 \ .20 \ .15 \ .10$ sum = 1.00 ✓



(a) at most 2	$P(X \leq 2) = .05+.10+.15 = .30$		
(b) at least 3	$P(X \geq 3) = 1 - P(X \leq 2) = 1 - .30 = .70$		
or $.25+.20+.15+.10 = .70$			
(c) between 2 and 5 inclusive	$P(2 \leq X \leq 5) = .15+.25+.20+.15 = .75$		
THE cdf-arithmetic habit:			
"at most k"	=	$F(k)$	add up to k
"at least k"	=	$1 - F(k-1)$	for an integer-valued X
"between a,b" =	$F(b) - F(a-1)$	for integer-valued X and integer a,b	

3. Worked: the find-k table (p066) - the batch-1 assignment long Q2

$x : 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7$
 $P(x) : 0 \ k \ 2k \ 2k \ 3k \ k^2 \ 2k^2 \ 7k^2 + k$

(i) FIND k

$$\sum p(x) = k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k$$

$$= 9k + 10k^2$$

$$10k^2 + 9k = 1 \rightarrow 10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0 \rightarrow (k+1)(10k-1) = 0$$

$k = 1/10$ (or -1 , rejected: probability cannot be negative)

(ii) with $k = 1/10$:

$$P(X < 6) = p(0) + p(1) + p(2) + p(3) + p(4) + p(5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= 0.80 + 0.01 = 0.81$$

$$P(X \geq 6) = 1 - 0.81 = 0.19$$

$$P(0 < X < 5) = p(1) + p(2) + p(3) + p(4) = k + 2k + 2k + 3k = 8k = 0.80$$

quick consistency check: $P(X < 6) + P(X \geq 6) = 0.81 + 0.19 = 1.00 \checkmark$

(iii) DISTRIBUTION FUNCTION: accumulate	
$F(x) = 0$	$x < 1$
0.1	$1 \leq x < 2$
0.3	$2 \leq x < 3$
0.5	$3 \leq x < 4$
0.8	$4 \leq x < 5$
0.81	$5 \leq x < 6$
0.83	$6 \leq x < 7$
1.0	$x \geq 7$

(iv) minimum a with $F(a) > 1/2$:

$$F(4) = 0.8 > 0.5 \rightarrow a = 4$$

$(F(3) = 0.5 \text{ is NOT } > 0.5, \text{ strict inequality})$

WATCH THE $P(x)=0$ ROW

The batch-1 edition has $P(X=0)=0$, giving $10k^2+9k=1 \rightarrow k=1/10$.

The SECOND edition changed $P(X=0)$ to k , giving $12k+10k^2=1$

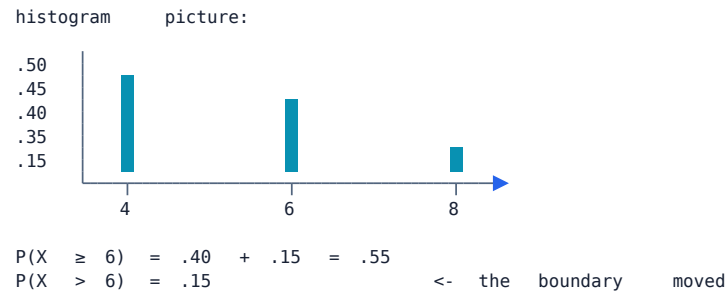
$\rightarrow k=0.0782$, but its own key still prints $k=1/10$ which does

NOT satisfy its own table. The second edition is broken

(errata 5.1). Study the batch-1 edition.

4. Worked: tune-ups, cylinders (p067-p069)

45% four-cylinder, 40% six-cylinder, 15% eight-cylinder				
X = number of cylinders of the next car				
x	:	4	6	8
$p(x)$: .45	.40	.15	sum 1.00 $\checkmark$	



5. Worked: building permits, proportional pmf (p070-p071)

Y = number of forms required, $p(y) = k y$, $y = 1, \dots, 5$

(a) $\sum p(y) = k(1+2+3+4+5) = 15k = 1$	->	$k = 1/15$
(b) $P(Y \leq 3) = (1+2+3)/15 = 6/15 = 0.4$		
(c) $P(2 \leq Y \leq 4) = (2+3+4)/15 = 9/15 = 0.6$		
(d) is $p(y) = y^2/50$ a pmf?	$\Sigma = (1+4+9+16+25)/50 = 55/50 \neq 1$	-> NO

THE VALID-PMF TEST IS ALWAYS: sum over the support = 1 ?
If it does not sum to 1, it is not a pmf. Period.

6. Worked: flashlight batteries (p072-p073) - a pmf built from a process

90% of batteries acceptable (A), need 2 acceptable to work.
 Y = number of batteries tested.

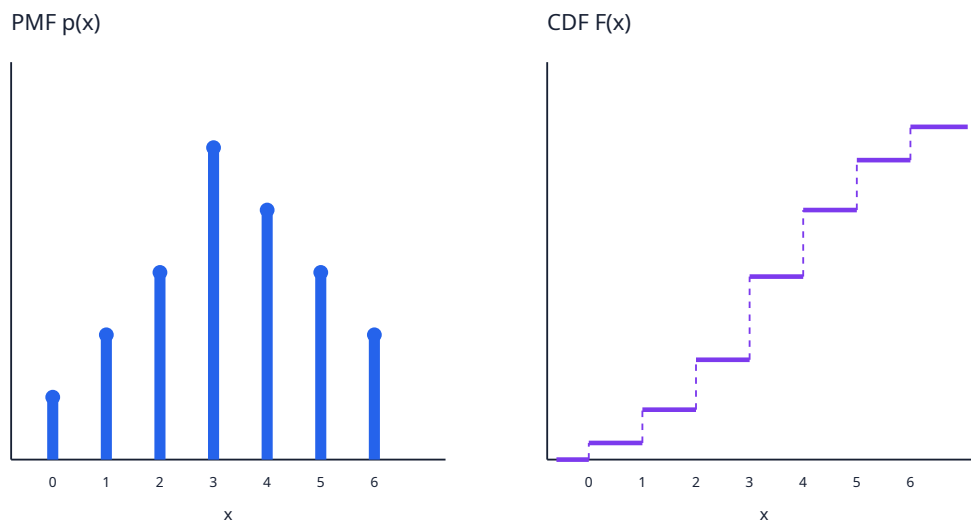
- (a) $p(2) = P(AA) = (.9)(.9) = .81$
 (b) $p(3) = P(UAA \text{ or } AUA) = (.1)(.9)(.9) + (.9)(.1)(.9)$
 $= 2(.1)(.9)^2 = .162$
 (c) $p(5)$: the 5th must be A, exactly one of the first four also A
 AUUUA, UAUUA, UUAUA, UUUAA = 4 outcomes
 $p(5) = 4(.1)^3(.9)^2 = .00324$
 (d) general: $P(Y=y) = (y-1)(.1)^{(y-2)}(.9)^2$, $y = 2, 3, 4, \dots$

The tree for part (b):

start		
/	\	
A(.9)	U(.1)	
A(.9)	A(.9)	<- UAA
done	A(.9)	<- UAA done
p(.81)		
done (AUA) $.9 \cdot .1 \cdot .9 = .081$		
total for $y=3$: $.081 + .081 = .162$		

7. The cumulative distribution function, cdf (p074-p078)

PMF accumulates into a CDF



How a PMF accumulates into a CDF

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y) \text{ (discrete)}$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy \text{ (continuous)}$$

PROPERTIES OF F

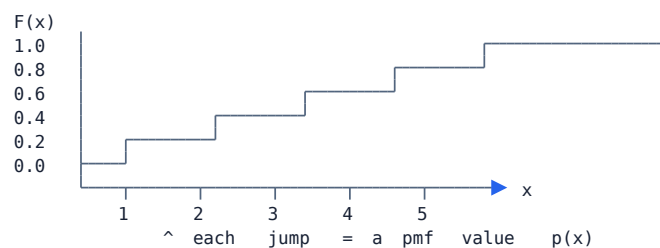
non-decreasing (it only accumulates)

$$0 \leq F(x) \leq 1$$

$$F(-\infty) = 0, F(+\infty) = 1$$

right-continuous

THE STEP-FUNCTION PICTURE (discrete cdf)



pmf AND cdf DETERMINE EACH OTHER

discrete: $p(x) = F(x) - F(x-)$ (the JUMP at x)

integer rv: $p(n) = F(n) - F(n-1)$

continuous: $f(x) = F'(x)$ (derivative)

8. THE key cdf theorem (p079)

for $a < b$:

$$P(a < X \leq b) = F(b) - F(a)$$

and the integer-valued version that catches people (p135):

$$P(a \leq X \leq b) = F(b) - F(a-1)$$

when X is integer-valued and a, b are integers

^ ^

the $a-1$ because $F(a-1)$ INCLUDES the value $a-1$, and

$P(a \leq X)$ must EXCLUDE everything below a

picture:

$F(b)$ = area /include everything up to and INCLUDING b
 $F(a)$ = everything up to and including a <- but $P(a < X)$ excludes a
 $F(a-1)$ = everything up to $a-1$ <- so this is the right one

... $a-1$ | a | ... | b
 └───┬───┘ └──┬──┘ └──┬──┘
 drop this keep keep

9. Worked: boards inspection (p080-p081)

Lots of 5 boards, 2 inspected. Boards 1,2 are the only defectives.

X = number of defective boards among the 2 inspected.

(a) all 10 selections: (1,2)(1,3)(1,4)(1,5)(2,3)(2,4)(2,5)(3,4)(3,5)(4,5)

$$(b) P(X=0) = P\{(3,4)(3,5)(4,5)\} = 3/10 = .3$$
$$P(X=2) = P\{(1,2)\} = 1/10 = .1$$
$$P(X=1) = 1 - (.3 + .1) = .6$$

(c) $F(0) = .30$

$$F(1) = .90$$
$$F(2) = 1.00$$
$$F(x) = 0 \quad x < 0$$

.30 $0 \leq x < 1$

.90 $1 \leq x < 2$

 $1 \leq x$

10. Worked: dice maximum M (p082-p083) - the elegant one

$M = \max$ of two dice

pmf: $p(1)=1/36$, $p(2)=3/36$, $p(3)=5/36$, $p(4)=7/36$, $p(5)=9/36$, $p(6)=11/36$

WHY the odd numbers? $M \leq m$ iff BOTH dice $\leq m \rightarrow m^2$ outcomes

$$p(M = m) = [m^2 - (m-1)^2]/36 = (2m-1)/36$$

check $m=3$: $(2*3-1)/36 = 5/36 \checkmark$

cdf:

$F(m) = 0$	$m < 1$
1/36	$1 \leq m < 2$
4/36	$2 \leq m < 3$
9/36	$3 \leq m < 4$
16/36	$4 \leq m < 5$

cdf:

25/36

1

^ at integer support points $m=1,\dots,6$, $F(m)=m^2/36$.

For real $1\leq m<6$, $F(m)=\text{floor}(m)^2/36$, as the steps above show.

m	F(m)
0	0
1	1/36 ≈ 0.03
2	4/36 ≈ 0.11
3	9/36 = 0.25
4	16/36 ≈ 0.44
5	25/36 ≈ 0.69
6	36/36 = 1.0

11. Worked: cdf is non-decreasing (p085) - the proof question

SHOW: $x_1 < x_2$	$\rightarrow$	$F(x_1) \leq F(x_2)$
proof:		
$F(x_2) = P(X \leq x_2)$		
$= P(\{X \leq x_1\} \cup \{x_1 < X \leq x_2\})$	split the event	
$= P(X \leq x_1) + P(x_1 < X \leq x_2)$	disjoint, so add	
$\geq P(X \leq x_1)$	a probability is ≥ 0	
$= F(x_1)$	✓	
AND: $F(x_1) = F(x_2)$ exactly when $P(x_1 < X \leq x_2) = 0$ (the interval between them carries no probability)		

pmf and cdf are done. Next: expectation and variance (notes p086-p111).